

Performance & Accuracy Testing Report

VANGUARD CREDIT AUTOMATION | ACADEMIC PROJECT TRACK DELIVERABLE

1. Algorithm Accuracy Comparison

- Decision Tree: Accuracy = 65.20% (Winner ■)
- XGBoost: Accuracy = 65.00%
- Random Forest: Accuracy = 64.80%
- Logistic Regression: Accuracy = 64.60%

2. REST API Load Performance

Predict API endpoint (/api/predict) yields response times under 50ms locally. Batch scans (/api/predict_batch) process 100-record CSV sheets in under 120ms.